

Experiment 7

Question

Capture video from a Web Camera and display the video in slow motion and in fast motion.

Aim

To capture live video from a webcam using Python and OpenCV and display it in slow motion and fast motion.

Procedure

1. Import the OpenCV (cv2) library.
2. Open the webcam using `cv2.VideoCapture(0)`.
3. Check whether the webcam is opened successfully.
4. Capture the video frame by frame.
5. Resize each frame for better display.
6. Display the video.
7. Use `cv2.waitKey(100)` for slow motion & `cv2.waitKey(10)` for fast motion.
8. Press 'q' to stop video and save output video

Code (Slow Motion)

```
import cv2

# Open webcam
camera = cv2.VideoCapture(0)

# Check webcam
if not camera.isOpened():
    print("Webcam not found!")
else:
    while True:
        ret, frame = camera.read()
        if not ret:
            break

        # Resize the frame
        frame = cv2.resize(frame, (640, 480))

        # Display webcam
        cv2.imshow("Slow Motion Webcam", frame)

        # Slow motion
        if cv2.waitKey(100) & 0xFF == ord('q'):
            break

    camera.release()
    cv2.destroyAllWindows()
```

Code (Fast Motion)

```
import cv2

# Open webcam
camera = cv2.VideoCapture(0)

# Check webcam
if not camera.isOpened():
    print("Webcam not found!")
else:
    while True:
        ret, frame = camera.read()

        if not ret:
            break

        # Resize the frame
        frame = cv2.resize(frame, (640, 480))

        # Display webcam
        cv2.imshow("Fast Motion Webcam", frame)

        # Fast motion
        if cv2.waitKey(10) & 0xFF == ord('q'):
            break

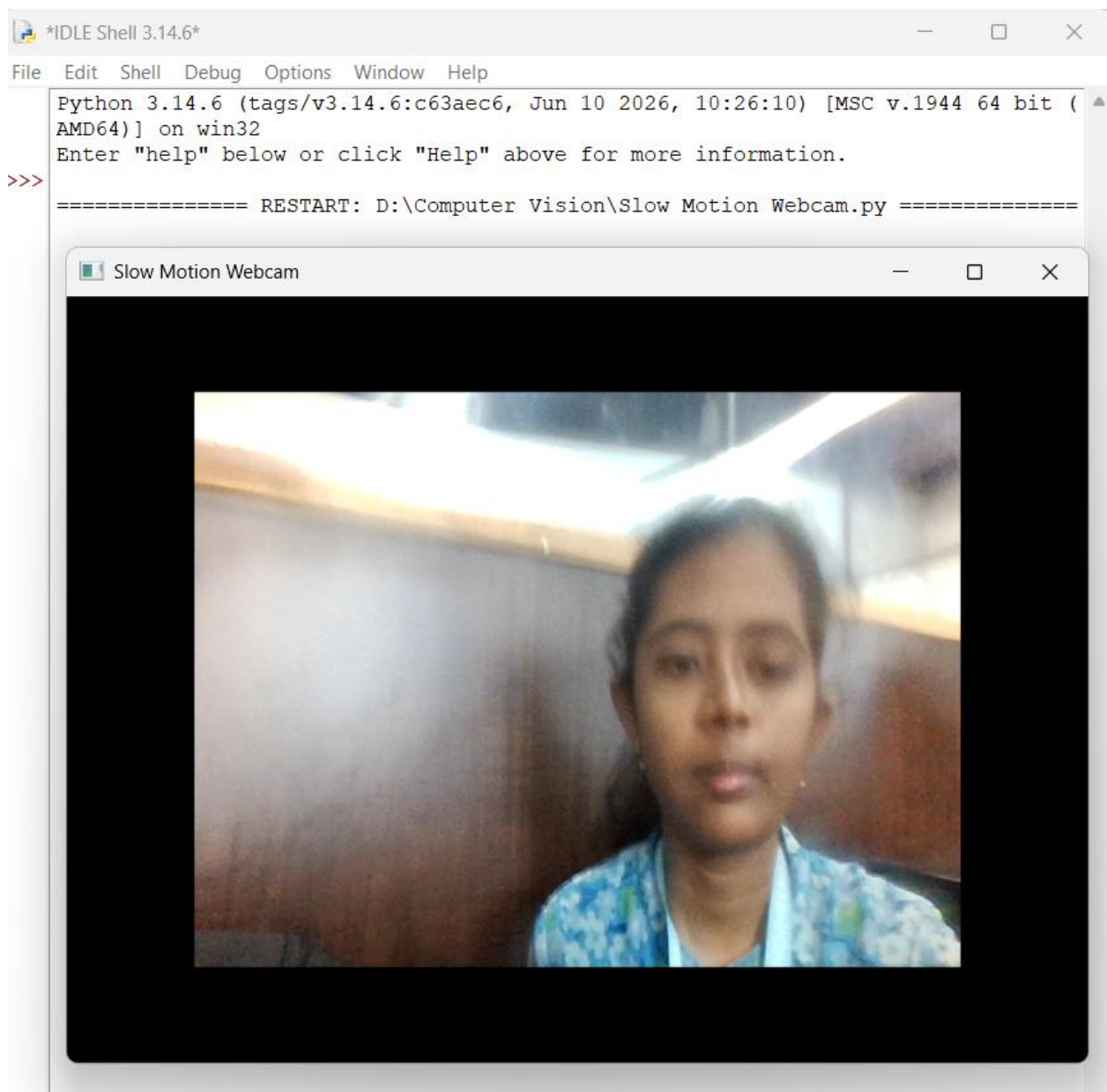
    camera.release()
    cv2.destroyAllWindows()
```

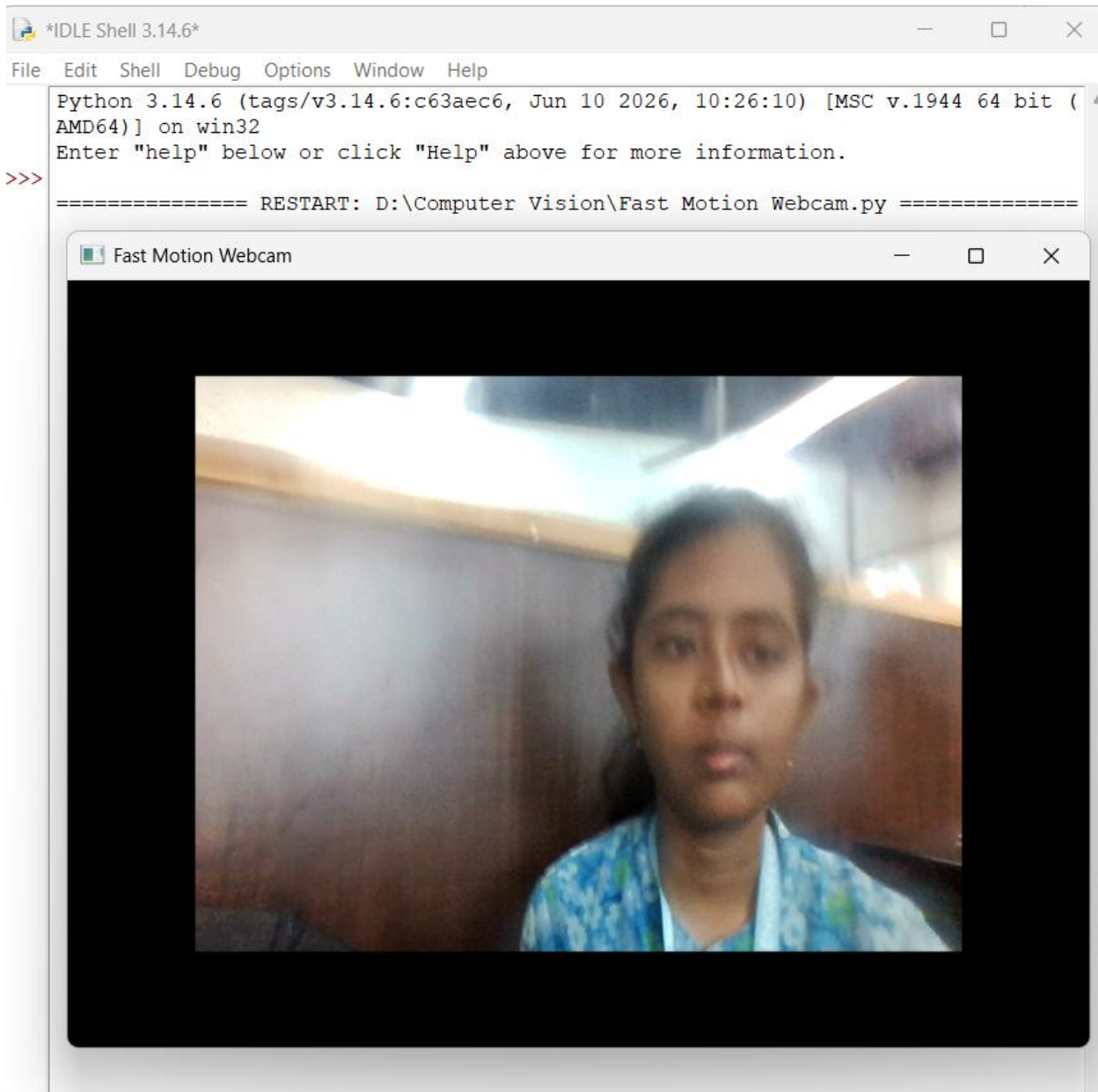
Input

Input Device: Laptop Webcam

Output

1. **Slow Motion Webcam** : Displays the live webcam video at a slower speed.
2. **Fast Motion Webcam** : Displays the live webcam video at a faster speed.





Result

The live video was successfully captured from the webcam using OpenCV and displayed in both slow motion and fast motion.